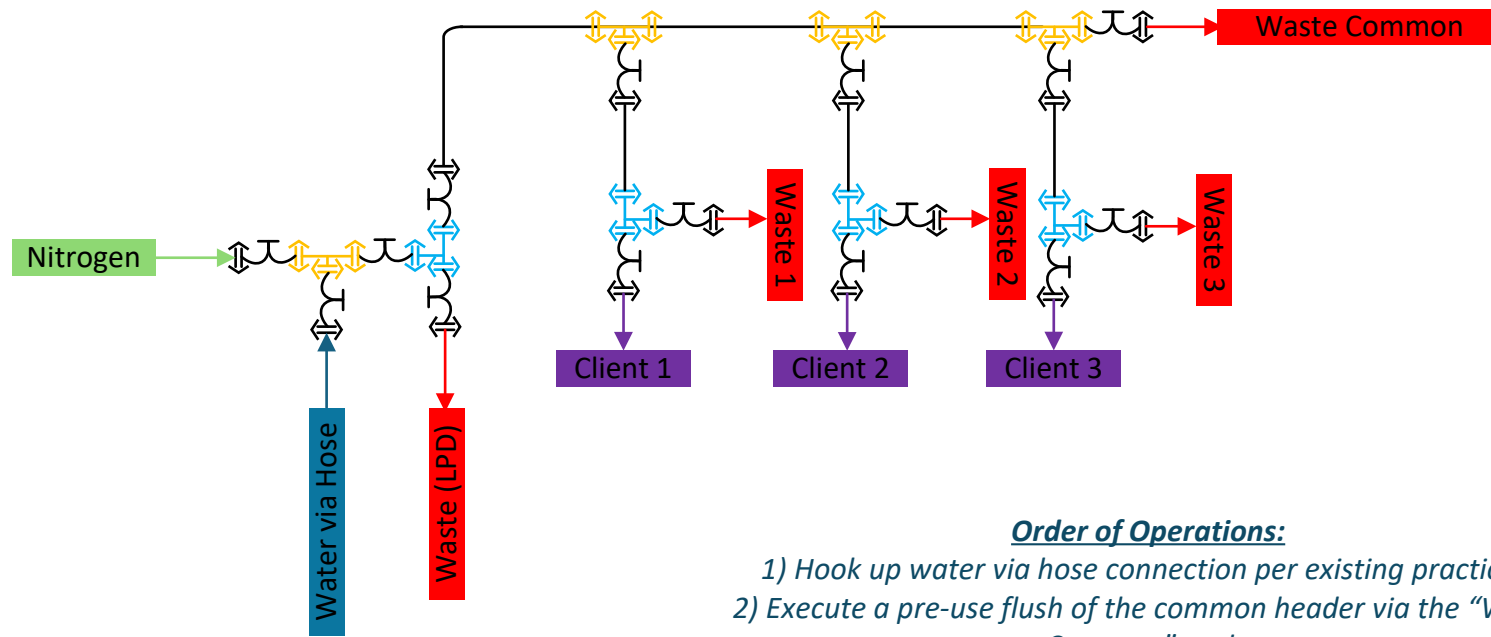
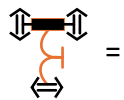
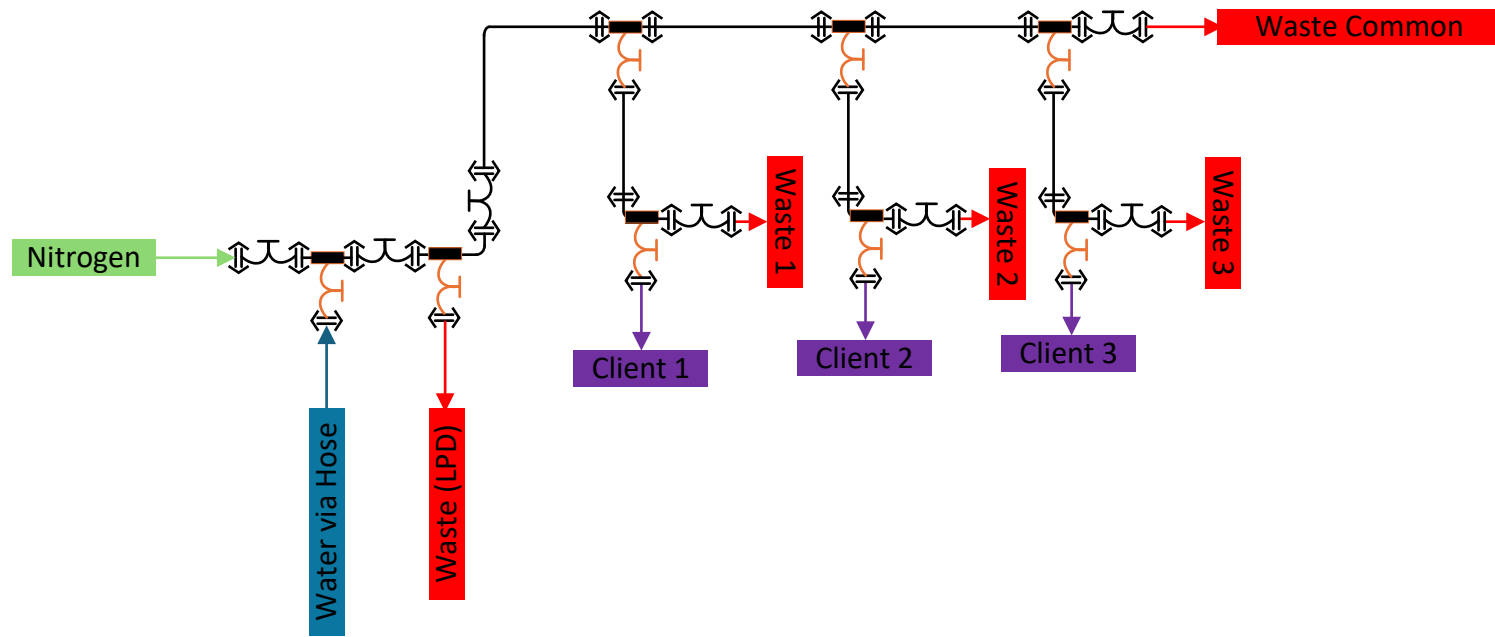




Note: Due to space limitations to fit clamps and overcome weld transition zones, only a leg from the run (top image, blue figure) or the branch (bottom image, orange figure) can be minimum distance, not both. As such, you need to thoughtfully place the minimum distance leg.

Order of Operations:

- 1) Hook up water via hose connection per existing practices.
- 2) Execute a pre-use flush of the common header via the "Waste Common" path.
- 3) Close off the "Waste Common" path and open path to client waste path. For example, if supplying "Client 1", open path to "Waste 1".
- 4) Execute a pre-use flush of the client specific header via the client waste path.
- 5) Close off client waste path and open path to client to supply desired amount of water. At end of addition, close client supply path and close water supply.
- 6) Disconnect/drain/stores water supply hose per existing practices.
- 7) Execute nitrogen blow of common header to "Waste Common" path. Then, execute nitrogen blow of client header to client waste path. Lastly, conduct a nitrogen blow to the LPD path. Close off nitrogen supply.

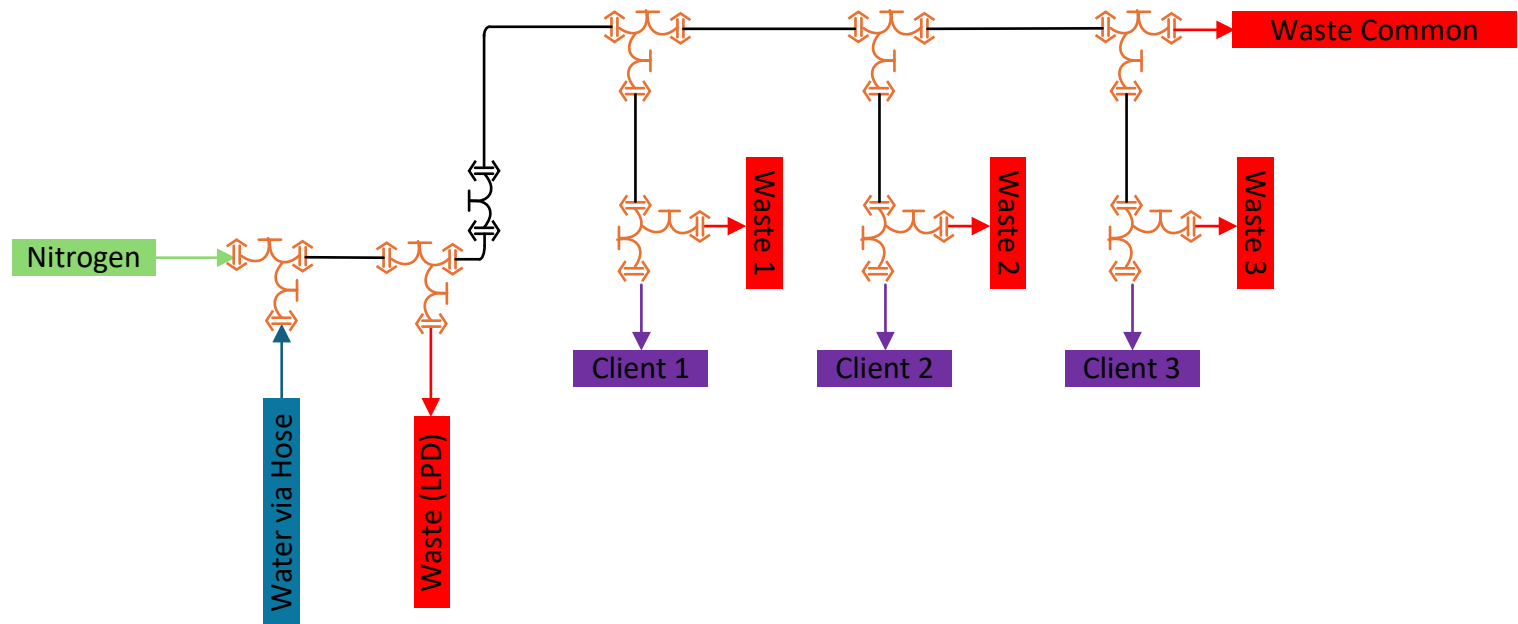


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<https://www.engvalves.com/Products/Zero-Static-Block-Body/>

Note: Zero-static or “zero dead leg” valves replace minimum distance tee with standard valve.



[https://www.engvalves.com/Products/Integral-Sterile-Access-and-GMP-\(ISG\)/](https://www.engvalves.com/Products/Integral-Sterile-Access-and-GMP-(ISG)/)

Note: ISG valve comes in several orientations/ configurations, so pay close attention to what is required based on field implementation. This provides zero static functionality in two directions and further segregates clients.